

Original Article

AR-Based Interactive Learning Application

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Abstract – The current project introduces an interactive application based on Augmented Reality (AR) and is created to improve user engagement by incorporating virtual reality content with the actual surroundings. Augmented Reality (AR) is transforming education by providing interactive and immersive learning experiences [1]. The app takes advantage of the camera, sensors, and AR systems to identify surfaces and objects or markers on a physical object and superimpose them with intuitive 3D visuals, animations, and interactive features. The system is made to be very realistic and dynamic by enabling users to interact in a natural manner, such as through touch, gesture, and movement. The primary objective of the application is to enhance the process of learning, exploring, or executing activities by users, making the information visually enhanced and readily understandable. The AR interface can be applied in education, training, entertainment, and product visualization, and is based on real-time feedback and promotes active interaction.

Keywords – Augmented Reality (AR), Interactive Application, 3D Content Overlay, Object Recognition, Interactive Learning, Mobile AR, Marker-Based Tracking, Human-Computer Interaction (HCI), Digital-Augmented Environment, Enhanced User Experience (UX).

1. INTRODUCTION:

With augmented reality (AR) becoming a more popular phenomenon and smartphones becoming more powerful, AR learning apps are beginning to shape the educational field significantly. It is now possible for students to use their phones to point at objects and view 3D models of those objects, play

in a virtual environment, engage with lessons, and learn in a more real and practical way. AR is one of the top emerging technology innovations in interactive media.[4]

Augmented Reality is a revolutionary technology that overlays digital information, such as images, audio, and text, on the real world, producing a combined immersive experience for the user. Within an educational setting, AR has the potential to transform student experience with learning materials entirely, providing interactive and highly immersive experiences that are far beyond what can be achieved with traditional textbooks, lectures, or even video.[1] Feeling frustrated because the app is not scanning correctly or tracking movements.

The potential of AR in education is vast, spanning a wide array of subjects, including history, mathematics, science, and engineering. Using AR applications, students can visualize complex concepts, explore historical events, and conduct virtual scientific experiments. These immersive experiences not only increase student engagement but also encourage a deeper understanding and retention of information. Additionally, AR promotes active learning by making education more interactive and hands-on, thus allowing students to engage with learning material in innovative ways.[7] intuitive design, prompts, guided interactions, etc., to make the experience smoother, easier to use, and learn.

2. LITERATURE REVIEW:

The system links three major categories: AR tracking technology, the management of 3D interactivity used for learning content, and the safety and functioning necessary for a mobile app. In short, it is divided into three categories: AR tracking and identification of real-world surfaces, the learning content management system within the 3D interface of the app itself, and the safety and functioning necessities of a mobile app.

2.1 AR Tracking and Recognition Techniques

In simple terms, AR spots things in the real world using a phone's camera and sensors and then places 3D models or animations on top of them. There are different types of AR detection methods, such as plane detection (finding floors and tables), image tracking (detecting pictures or markers), object tracking (recognizing shapes), depth sensing, and even motion patterns from the device. Among these, image tracking is the most familiar because it only requires regular camera pictures, making it easy and natural for users.

The system captures user interactions, such as taking photographs of instructional materials, using handheld devices. The images were subsequently processed in multiple stages for effective recognition and

categorization. [1]

The arrival of the Viola–Jones method accelerated image detection. It uses small, quick filters to look through tiny portions of the image one by one. Instead of heavy calculations, it quickly skipped regions that did not match, allowing real-time detection even for older phones and cameras. Because of this speed and simplicity, the technique has become common in many applications, such as camera filters, barcode scanners, educational apps, and simple AR tools.

Soon, methods such as LBP began to improve recognition by focusing on the texture around each pixel. These methods handled changes in lighting much better than earlier methods and worked faster.

As AR technology grew, deep learning systems, largely similar to Face Net but now targeted at AR-related tasks, learned how to transform real-world objects or images into detailed digital representations called embedding. Trained on large collections of images, these models can place pictures of the same object close together and separate different objects far apart. This finally made AR tracking much more stable, accurate, and reliable, even in challenging environments. These advancements have continued to make today's AR apps much more capable, not only in detecting surfaces and recognizing markers but also in smoothly placing 3D content so that users can naturally interact with this digital world. AR works by using a phone's camera and sensors to understand the real world and correctly place digital objects. Early AR systems used simple patterns of images, but they were confused by changes in lighting, angles, and movement. Techniques such as Viola-Jones improved speeds by using small, fast filters that efficiently scanned images and made real-time detection possible on even older phones. Methods such as LBP handle variations in lighting much better because their basic approach concentrates on the texture around each pixel.

The practical use of AR in schools includes AR textbooks, mobile applications, and wearable devices, such as smart glasses. Educators are integrating AR into science experiments, historical reconstructions, and language learning.[1] Today's AR frameworks, such as AR Core, combine plane detection, motion tracking (SLAM), depth sensing, and image recognition to create stable, smooth, and realistic AR experiences.

2.2 AR Data Security and Content Protection

Cryptography is a major factor that helps secure AR app data. For AR learning apps, it includes safeguarding the following information: user progress, stored files of the lessons learned, 3D models, and sensitive information that the app possesses.

AR applications today involve robust encryption algorithms, in the same manner that AES encryption is used in learning institutions and organizations to secure digital content. AES-256 encryption can be considered trustworthy because it is difficult for hackers to crack due to its long key size when

used to store critical learning content or user data offline. However, the ability to protect these encryption keys makes any secure system robust. If such keys are derived from weak passwords or stored in plain sight, it will be easy for hackers to guess the keys and steal the offline versions. Even with the use of tools such as PBKDF2, BCrypt, or SCrypt, security still lies in the password strength.

For Android platforms, the Android Key store system is available to developers as a system security component. It replaces the requirement to allow apps to store key mappings in a file and enables the generation of key mappings within a locked area, either in a dedicated hardware component or deep within the operating system itself. In this system, the key does not leave the device, and unlocking is made possible through PIN or fingerprint verification.

For AR apps, this means:

- The files containing 3D content remain protected
- User progress and saved sessions are kept private
- Offline AR experiences are always safe, no matter who duplicates the app's content

However, even if data are stolen through the AR app, it would be very difficult to gain access to the encryption keys. As a consequence of these keys being stored within the Android Keystore system, there would be much greater hurdles for hackers, thereby ensuring a safer and more trustworthy AR learning environment for the users. Within the AR learning app, cryptography ensures that data belonging to users, as well as 3D models and lesson files, remain secure from malicious activities. AES-256 encryption methods would be ideal as they are very difficult for hackers to crack. However, this largely depends on the encryption keys generated from passwords or open files accessible by hackers.

The system captures user interactions, such as taking photos of instructional materials, through handheld devices. The images are subsequently processed in multiple stages for effective recognition and categorization.[1]

4.3 Mobile Application Security and Authentication in AR Apps

The aim of mobile AR app security is to safeguard the user and their data while allowing them to engage seamlessly with digital content. Mobile apps might contain users' progress information and other graphics and files related to teaching; therefore, it is essential to safeguard this data.

Authenticating users using passwords, tokens, or biometric measures has its advantages and disadvantages.

- Although passwords are easy to create, they are often weak.
- Biometric entry through fingerprints or facial recognition increases convenience; however, spoofing biometrics or privacy concerns may still pose a threat.

Experts have pointed to the typical risks associated with mobile applications.

- Storing confidential data either unencrypted or with poor encryption
- Having apps/components with excessive permissions to steal or display information to malicious software
- Using insecure network connections without proper safeguards

The best practices for AR learning apps are as follows:

- Encrypting all data - all saved files, records of progress, and AR models
- Leaning on available security features instead of implementing ad hoc safeguards
- Supporting local operations for sensitive functions and minimizing cloud dependencies
- It seems that a secure AR app can be based on these principles by integrating the following:
- Biometric login systems (Facial recognition login) for easy access
- Storing 3D content and lesson data using strong encryption
- Offline functionality, preventing unnecessary online use

With these steps in mind, AR-enabled applications can ensure an immersive learning experience while preserving users' private data in a secure and fully controlled manner on the device.

3. PROBLEM DEFINITION/ RESEARCH GAP

Conventional learning applications have attempted to simplify the learning process; however, new issues tend to arise with them. There are numerous AR-based applications that provide 3D models, animations, and interactivity; however, they depend on an elaborate menu, confusing controls, or regular scanning and tracking. Unless the system is developed properly, the overall learning process will rely on the ease with which a user can engage with AR features. Students lose interest when the interface is not easy to use or when tracking fails.

Students can now visualize complex or challenging subjects in ways that were previously unthinkable, thanks to AR. For instance, students may use AR devices to engage with 3D simulations of solar systems or internal body structures instead of reading about human anatomy or planetary motions. [1]

AR learning systems have quite a number of big problems that reoccur:

Overreliance on stable tracking: AR activities are overwhelming, particularly for new users. If students have no idea where to look to scan an object, place their camera, or engage with 3D elements, they become frustrated and leave the AR materials.

Over-reliance on smooth tracking: When the AR experience is pegged on the light being good, the scanning being accurate, or the movement of hands being consistent, a small mistake would disrupt the activity. In the case of tracking failure, the entire lesson is wasted.

Privacy issues and reliance on clouds: Some AR apps save user data, progress in the learning process, and even camera images on third-party servers. This poses privacy concerns, and students are unable to access some important features when offline.

Poor utilization of device capabilities: Not every AR learning application utilizes phone-inbuilt sensors, local processing, and offline functionalities to the fullest. Others overly use cloud-based services or third-party libraries, which makes them slower and less accurate.

Educational content becomes more effective when teachers utilize AR features together. Students become more engaged when complex abstract subjects become interactive hands-on experiences that also accommodate different learning preferences and support active experiential learning.[8]

4. OBJECTIVE OF THE STUDY

The key aim of this research is to design and develop an AR learning application for Android that enables a more interactive and easily comprehensible learning process with the help of AR rather than the normal learning process taught through static concepts. This research also includes an analysis of AR-based educational tools and games that are currently available for children. These tools were assessed for their effectiveness in promoting engagement, creativity, and critical thinking in different subjects. Additionally, the study considers feedback from educators, parents, and children to gauge the real-world impact of AR technologies on children's learning experiences.[7]

AR also promotes collaborative learning by allowing students to share the same augmented experiences and participate in group problem-solving activities, making learning more dynamic and interactive.[6] In this section, we summarize the relevant research on AR for second language learning and mobile learning in general. We highlight the theoretical support and empirical evidence of the impact of contextual tools and AR in this domain. [3] For scenarios and contexts in which the tracking within the augmented reality experience may not be feasible—for instance in low-light environments or in environments with reflective surfaces as well as situations in where the camera within the device might be malfunctioning or inadequate—the app will provide fallback alternatives such as simplified preview or explanation in the form of texts to ensure a seamless experience in conducting a particular or specified 'lesson.'

5. COMPARATIVE ANALYSIS:

The use of AR in learning spaces also accommodates central pedagogical ideals, such as active learning and constructivist models.[1]

While analyzing various ways of managing content delivery and interaction through AR learning application software, I found that no two approaches were similar in their impact on feelings of control and trust.

This study adopts a qualitative research methodology, focusing on a review of the existing literature and case studies to explore the role of AR in interactive learning for children. A variety of sources were consulted, including academic journal articles, educational reports and industry publications. These sources provided insights into the use of AR in different educational contexts, including schools, museums, and home-based learning environments.[7] However, to perform a comprehensive evaluation or synthesis of findings on a specific topic, an SLR is more appropriate. An SLR not only helps pinpoint future research directions but also addresses existing research gaps. Because it involves an in-depth examination of all relevant studies conducted on a topic, it requires considerable time and effort. Nevertheless, adopting a robust research methodology can significantly enhance the quality and thoroughness of the SLR.[8]

One thing that I enjoy most about offline device-based AR applications is the integration of both new interactivity and the end-user's control as a key component in AR-driven learning environments. These systems allow learners to engage with 3D lessons, scan objects, and access AR functionalities without necessarily going to the cloud environment to gain access to these services.

There are some considerations for each method.

- The classical marker-based AR system may appear to be restrictive and error-prone.
- Cloud-based AR offers ease of use but requires reliance on a third party and a stable Internet connection.
- AR applications running locally and offline enable high-speed and smooth performance with safety and content control in the learner's hands.

The offline, device-based solution appears to be the optimal choice, providing an immersive, interactive educational experience that is user-friendly, reliable, and private.

5.1 Comparison:

1. Where Content and Data Are Stored

- Often, traditional AR app storage is conducted either on a device or via cloud storage solutions with limited space.

- Cloud-based AR platforms allow the storage of all content and user data on remote servers that belong to another person.
- The new AR application provides all lessons, models, and any progress made locally on the user's device and does not upload or share any content.

2. How Data Security Keys Are Managed

- More outdated AR systems can relate encryption directly to passwords, where security is dependent on the code.
- Cloud AR platforms may allow their servers to control the keys. Therefore, your key security is in their hands.
- In this AR app, a strong key for encryption has been developed and stored in a secured system using Android Keystore and never goes out of your mobile device.

3. Internet Requirement

- A classical AR app can typically operate without the Internet.
- Cloud-based AR apps require an internet connection; otherwise, loading of content or access to progress may not be possible.
- The innovative app using AR technology operates entirely in offline mode; therefore, all learning and interactive models are available to the user.

4. Handling Biometric Data

- The majority of older AR systems do not use biometrics.
- Cloud services may store fingerprints or facial information online, which poses a privacy risk.
- The AR app will store your face or fingerprint information only on your phone, and no one else will be able to access it.

5. Risk from Weak Access Codes

- Password-based traditional AR applications can be insecure if the password is weak.
- Cloud infrastructure should have robust login credentials; otherwise, it may affect the data.

- The new AR system uses hardware-backed keys and extra protection; therefore, weak PINs or codes are less dangerous.

6. Security in Case Your Device Is Lost or Stolen

- Older AR apps rely on passwords alone; a thief could gain access if the password is weak.
- Cloud AR platforms may allow users to lock their accounts remotely, but local content may remain more vulnerable to exposure.
- The proposed AR app requires bypassing face or PIN authentication coupled with Keystore hardware protection, which makes unauthorized access difficult.

7. User Convenience

- Older augmented reality systems can be clunky and require repeated logins or other setup measures.
- Cloud AR apps are convenient if the Internet is strong and the servers are working.
- Fast and Easy: The new AR app requires only the scanning of one's face to start learning anywhere and at any time.

8. Privacy and Trust

- Traditional AR apps require trust in the app and syncing services.
- Cloud AR platforms require users to trust the provider with all their data.
- The new approach of AR does not share anything but keeps everything private on the device, and the user remains in full control.

6. PROPOSED WORK OVERVIEW:

The solution to the intended AR-based educational system is constructed by blending interactive AR models for instructional purposes. The system captures user interactions, such as taking photos of instructional materials, through handheld devices. The images are subsequently processed in multiple stages for effective recognition and categorization.[1]

The results of this study demonstrate the potential of AR to revolutionize interactive learning for children. AR applications provide an engaging, creative, and immersive way to learn, making abstract concepts tangible and accessible. By fostering critical thinking, creativity, and collaboration, AR helps develop essential skills that are crucial for children's cognitive and social development.[7] This way, you can see how the app turns normal lessons into an interactive and engaging AR experience.

6.1 System Architecture

AR-Based Interactive Learning Application

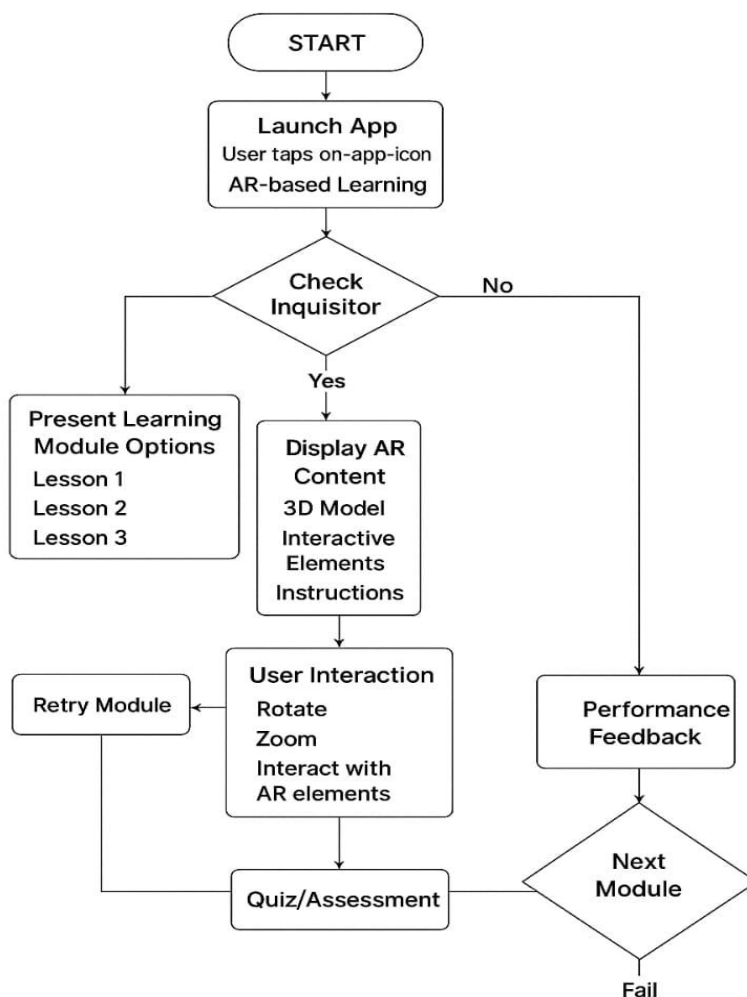


Figure 1: AR-Based Interactive Learning Application

1) Presentation Layer (AR User Interface)

This is the part with which learners interact directly.

- First, display AR lessons, 3D models, quizzes, and progress dashboards.
- Step-by-step guides the user through starting a lesson, selecting objects to explore, or viewing interactive animations.
- It handles gestures, taps, or voice commands for navigation.
- The input is sent to the AR engine and security layers without any background noise.

2) AR Tracking & Object Recognition Layer

This layer allows the app to “see” the real world.

- Discriminates between surface, marker, or object (e.g., a textbook page or physical model).
- Maps 3D content onto real-world locations so that virtual objects appear stationary and interactive.
- It tracks the user's movement and updates AR content in real time, allowing for fluid experiences.
- Face recognition (optional) for secure logins or personalized lessons.

3) Security & Authentication Layer

Keeps learner data and progress safe.

- It supports face recognition or a PIN for logging in to unlock the AR app.
- Creates strong encryption keys using the Android Keystore.
- It encrypts all sensitive data on the user's profile, progress, AR content, and notes.
- Ensuring that all authentication and sensitive operations occur locally and offline without data being sent to the cloud.

4) Data & Content Storage Layer

Manages all learning materials and progress:

- It stores 3D models, interactive content, and lesson data in a local database.
- Saves encrypted user data like facial code, backup PIN, and quiz scores.
- Keeps everything private and offline; nothing is stored as plain text or sent online.

5) Application Logic Layer (AR Engine & Learning Flow)

Coordinates the AR experience and user interaction:

- Interprets user input and translates it into actions (e.g., opening a 3D model or starting a quiz).
- Saves encrypted user data such as facial code, backup PIN, and quiz scores.
- Handles animations, physics, and object behaviors in the AR scene.
- It interfaces with the storage layer to load or save content where necessary.

6) First-Time Launch & Setup

- Verifies if a profile exists.
- If not, guides the user to scan their face or create a PIN.
- Provides local storage of login credentials in encrypted form.
- It prepares the local database for storing AR lessons, progress, and 3D assets.

7. CONCLUSION:

The AR-based interactive learning application demonstrates how augmented reality can substantially enrich learning by transforming abstract or complex ideas into entertaining, visual, and manipulative interactions. The integration of digital content and the real world within this application enhances interest among learners, provides them with better insights, and leads to increased knowledge retention compared to traditional methods of learning.

Overall, this AR-based interactive learning application appears to bring great potential toward modernizing education: making it more immersive, effective, and learner-cantered. With further enhancement and wider adoption, such applications can play a very significant role in the future of digital education.

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